ALLOY

STACKED MONO SYNTH — QUIC MANUAL

MIXF Inc — Project Volume
Version 0.3.8
2025 · v2 / 2273 / Synthesizer

ANALOG / FM PAGE

SUB / ARP / MIX PAGE

01 What It Is

Alloy is a monophonic bass synth built by stacking three voices into one: an analog-style oscillator with unison and its own resonant filter, a sub-oscillator underneath it, and a two-operator FM voice alongside. All three sum together before Drive and Tone shape the combined signal, with a built-in arpeggiator for sequenced patterns.

Age adds slow pitch drift and a faster wobble to the analog oscillator, modeling an old synth that's drifted out of calibration. At 0% (the default) it's off entirely. Turning it up adds progressively more wobble.

02 Signal Path

The Analog, Sub, and FM voices sum together before Drive, Tone, and Age shape the combined signal on its way to Output. The arpeggiator sequences notes rather than touching the audio path directly.

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NOTES
VOICES  ANA · SUB · FM
        → DRIVE / TONE → AGE → OUTPUT
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03 The Panel

Two pages share this panel. Analog and FM are shown by default; click the header button to switch to Sub, Arp and Mix, and back again.

ANALOG · FM CARRIER · FM MODULATOR · SUB · ARP · MIX

ANALOG

Waveform Saw / Square / Triangle · default Saw
The analog oscillator's waveform.

Unison 2 / 3 / 4 / 5 / 6 · default 3
Number of detuned voices stacked on the analog oscillator.

Octave -2 / -1 / 0 / 1 / 2 · default -1
The analog oscillator's octave transpose.

Detune 0-100% · default 20%
Spread between the unison voices. Has no effect at Unison 1.

Filter Cutoff 50 to 20 kHz · default 2200 Hz
Cutoff frequency of the analog voice's resonant filter.

Filter Resonance 0-100% · default 25%
Emphasis at the cutoff frequency. Higher values give the filter a more pronounced, whistling peak.

Filter Env Amount 0-100% · default 40%
How far the Filter Envelope sweeps the cutoff away from its knob position.

Vel→Filter 0-100% · default 40%
How much MIDI velocity opens the filter on harder-played notes.

Filter Envelope Default: 0.005 s / 0.100 s / 0% / 0.400 s
Shapes the cutoff sweep set by Filter Env Amount. Attack and Decay shape the initial sweep, Sustain holds it (0% lets the cutoff fall all the way back to its resting position), and Release shapes the sweep after note-off.

Amp Envelope Default: 0.005 s / 0.200 s / 80% / 0.300 s
Shapes the analog voice's loudness over the note's life. Attack and Decay shape the initial swell, Sustain holds the level while a note is held, and Release shapes the fade after note-off.

Glide 0-1 s · default 0 s
Portamento time between notes. At 0 s notes jump directly to pitch.

Volume 0-100% · default 60%
The analog voice's level in the final mix.

FM CARRIER

Waveform Sine / Triangle / Square · default Sine
The carrier oscillator's waveform.

Octave -2 / -1 / 0 / 1 / 2 · default -1
The carrier oscillator's octave transpose.

Volume 0-100% · default 70%
The carrier's level in the final mix.

Vel→Carrier 0-100% · default 70%
How much MIDI velocity affects the carrier's level on harder-played notes.

Envelope Default: 0.002 s / 0.300 s / 40% / 0.100 s
Shapes the carrier's loudness over the note's life, the same Attack/Decay/Sustain/Release shape as the Analog voice's Amp Envelope.

FM MODULATOR

Waveform Sine / Triangle / Square · default Sine
The modulator oscillator's waveform.

Octave -2 / -1 / 0 / 1 / 2 · default -2
The modulator oscillator's octave transpose.

Volume 0-100% · default 30%
The modulator's own level in the final mix, independent of how much it's modulating the carrier.

Vel→Bright 0-100% · default 50%
How much MIDI velocity adds to Brightness on harder-played notes.

Brightness 0-100% · default 0%
FM index: how strongly the modulator shifts the carrier's pitch each cycle. At 0% the carrier is unmodulated. Turning it up adds increasingly bright, bell-like or metallic overtones.

Envelope Default: 0.001 s / 0.150 s / 0% / 0.050 s
Shapes the modulator's own loudness over the note's life, the same Attack/Decay/Sustain/Release shape as the other envelopes. Since the modulator drives Brightness, this envelope also shapes how the FM tone evolves over the note.

SUB

On the Sub / Arp / Mix page.

On default: on
Turns the sub-oscillator on or off.

Waveform Square / Sine / Triangle · default Square
The sub-oscillator's waveform.

Octave -3 / -2 / -1 · default -2
The sub-oscillator's octave, relative to the analog voice.

Volume 0-100% · default 40%
The sub-oscillator's level in the final mix.

ARP

On the Sub / Arp / Mix page.

On default: off
Turns the arpeggiator on. While on, held notes are sequenced instead of played as a chord.

Sync default: on
Locks the arp's step rate to a tempo-synced note division (Division) instead of Rate's free Hz value.

Division 1/4 · default 1/16
The arp's step rate as a tempo-synced note division, while Sync is on.

Rate 0-20 Hz · default 4 Hz
The arp's step rate, free-running. Unused while Sync is on.

Pattern Up / Down / Up Down / Random / At Chord · default Up
The order notes in a held chord are played back in.

Oct. Range 1 / 2 / 3 / 4 · default 1
How many octaves the pattern spans before repeating.

Gate 0-100% · default 70%
How long each step's note stays on, as a percentage of the step length. Short values give a tight, staccato feel.

Hold default: off
Keeps the arpeggio playing after keys are released, until a new chord is played.

MIX

On the Sub / Arp / Mix page.

Drive 0-100% · default 20%
Adds saturation to the combined signal before Output.

Tone 0-100% · default 70%
Tilts the combined signal's tone: darker at low values, brighter at high.

Output -24 to +12 dB · default 0 dB
Final output trim.

Age 0-100% · default 0%
Adds slow pitch drift and a faster wobble to the analog oscillator, modeling an old synth that's drifted out of calibration. At 0% it's off entirely.

A **Panic** button sits at the top right of the panel. It force-stops all voices immediately, useful if a note gets stuck.

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